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4.1.1. Pandas - series creation and manipulation

00:32

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

Submit

12

grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]

13

print("Mean of even and odd numbers:")

14

print(grouped)

15

import pandas as pd

16

Average time

0.015 s

16.00 ms

Maximum time

0.041 s

41.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Mean of even and odd numbers:

Odd: ... 5.0

Even: ... 6.0

dtype: float64

Mean of even and odd numbers:

Odd: ... 5.0

Even: ... 6.0

dtype: float64

Test case 2

12 ms

Test case 3

9 ms

Terminal

Test cases

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8:01 PM

5/6/2026

ENG

4.1.2. Dictionary to dataframe

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

datafram...

```
1 # import pandas as pd
2
3 # # Provided dictionary of lists
4 # data = {
5 #     'Name': ['Alice', 'Bob', 'Charlie'],
6 #     'Age': [25, 30, 35],
```

Average time: 0.153 s, Maximum time: 0.159 s

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1 159 ms Debug

Expected output	Actual output
Original DataFrame:	Original DataFrame:
.....Name Age	Name Age
0 Alice 25	0 Alice 25
1 Bob 30	1 Bob 30
2 Charlie 35	2 Charlie 35
New name: Susan	New name: Susan

Terminal Test cases

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4.1.3. Student Information00:25

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

• Display the first five rows of the data frame.

• Calculate the average age of the students (limit the average age up to 2 decimal places).

• Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:
Refer to the displayed test cases for better understanding.

Sample Test Cases

studentin...

11

12# Read the text file into a DataFrame

13file = input()

14data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age", "Grade"])

15

Average time0.066 s66.00 ms

Maximum time0.092 s92.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 192 ms

Expected output

Actual output

studentdata.txt

studentdata.txt

First five rows:

First five rows:

0 John 25 A

0 John 25 A

1 Alin 22 B

1 Alin 22 B

2 Emma 24 A

2 Emma 24 A

TerminalTest cases

< Prev

Reset

Submit

Next >

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8:02 PM5/6/2026

4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

monthFor...

```
1 # import pandas as pd
2
3 ## Prompt the user for the file name
4 # file_name = input()
5
6 ## Load the data
```

Average time: 0.031 s (31.00 ms) Maximum time: 0.064 s (64.00 ms)

1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 (64 ms) Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
Best month: 2025-01	Best month: 2025-01
Total sales: \$1210.00	Total sales: \$1210.00

Terminal Test cases

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4.2.2. Best Selling Product00:23

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

monthFor... Submit

1# import pandas as pd

2

3## Prompt the user for the file name

4# file_name = input()

5

6## Load the data

Average time0.028 s27.67 ms

Maximum time0.058 s58.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 158 ms Debug

Expected output

sales_data.csv

Best-selling product: Product A

Total quantity sold: 23

Actual output

sales_data.csv

Best-selling product: Product A

Total quantity sold: 23

Terminal

Test cases

< Prev Reset Submit Next >

Type here to search

8:03 PM5/6/2026

4.2.3. City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:

monthFor...

```
1 # import pandas as pd
2
3 ## Prompt the user for the file name
4 # file_name = input()
5
6 ## Load the data
```

Average time: 0.029 s (29.33 ms) Maximum time: 0.061 s (61.00 ms)

1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 (61 ms) Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
City sold the most products: Los Angeles	City sold the most products: Los Angeles

Terminal Test cases

Prev Reset Submit Next

01:41 A [icon] [icon] [icon] [icon] [icon]

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Date	Product	Quantity	Price	City
2025-01-01	Product A	5,20	New York	
2025-01-01	Product B	3,15	Los Angeles	
2025-01-02	Product A	7,20	New York	
2025-01-02	Product C	4,30	Chicago	
2025-01-03	Product B	2,15	Chicago	
2025-01-03	Product A	8,20	Los Angeles	
2025-01-04	Product C	6,30	New York	
2025-01-04	Product B	5,15	Los Angeles	
2025-01-05	Product A	3,20	Chicago	
2025-01-05	Product C	10,30	Los Angeles	

Explanation:
Transactions:

Terminal Test cases

4.2.5. Titanic Dataset Analysis and Data Cleaning

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Sex	Age	SibSp	Parch	FamilySize	Fare	Cabin	Embarked
1	0	female	35	1	0	2	53.1	C55	S
2	1	male	29	0	0	1	52.0	C57	S
3	1	male	25	1	0	2	51.0	C59	S
4	1	male	35	1	0	2	51.0	C59	S
5	0	female	35	1	0	2	53.1	C55	S

titanicDat...

```
1 # import pandas as pd
2 # import numpy as np
3
4 ## Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6
```

Average time: 0.328 s (328.00 ms) Maximum time: 0.328 s (328.00 ms) 1 out of 1 shown test case(s) passed

Test case 1 (328 ms) Debug

Expected output	Actual output
PassengerId Survived Pclass Fare Cabin Embarked	PassengerId Survived Pclass Fare Cabin Embarked
0 1 0 3 7.25 S	0 1 0 3 7.25 S
1 2 1 71.28 C85	1 2 1 71.28 C85
2 3 1 7.92 S	2 3 1 7.92 S

Terminal Test cases

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Brown	male	22	1	0	12345	53.1	C123	S
2	1	1	Mr. White	male	35	0	0	67890	51.0	A987	C
3	0	3	Mr. Green	male	19	0	0	11223	33.5	F456	Q
4	1	2	Mr. Black	male	28	0	0	44556	36.0	D789	S
5	0	3	Mr. Grey	male	40	1	0	77889	47.5	E123	C
6	1	1	Mr. Red	male	30	0	0	99001	52.0	B567	Q
7	0	3	Mr. Blue	male	18	0	0	22334	31.0	G890	S
8	1	2	Mr. Yellow	male	25	0	0	55667	34.0	H234	C
9	0	3	Mr. Purple	male	38	1	0	88990	45.0	I567	Q
10	1	1	Mr. Brown	male	32	0	0	33445	50.0	J890	S

titanicDat...

```
1 # import pandas as pd
2 # import numpy as np
3
4 ## Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6 # data['FamilySize'] = data['SibSp'] + data['Parch']
```

Average time: 0.113 s (113.00 ms) Maximum time: 0.113 s (113.00 ms) 1 out of 1 shown test case(s) passed

Test case 1 (113 ms) Debug

Expected output	Actual output
29.69911764705882	29.69911764705882
14.4542	14.4542
3...491	3...491
1...216	1...216
2...184	2...184
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
0 677	0 677

Terminal Test cases

< Prev Reset Submit Next >

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
-------------	----------	--------	------	-----	-----	-------	-------	--------	------	-------	----------

titanicDat...

```
1 # import pandas as pd
2 # import numpy as np
3
4 ## Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6 # data['FamilySize'] = data['SibSp'] + data['Parch']
```

Average time: 0.188 s, Maximum time: 0.188 s, 1 out of 1 shown test case(s) passed

Test case 1 188 ms Debug

Expected output	Actual output
Pclass	Pclass
1 --- 0.629638	1 --- 0.629638
2 --- 0.472826	2 --- 0.472826
3 --- 0.242363	3 --- 0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S
0 0.629638	0 0.629638

Terminal Test cases

4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Braund	male	22	1	0	15160122	5.12	103	S
2	1	1	Mr. Allen	male	35	0	0	15160122	5.12	103	S
3	0	3	Mr. Brown	male	29	1	0	15160122	5.12	103	S
4	1	1	Mr. Davis	male	35	0	0	15160122	5.12	103	S
5	0	3	Mr. Miller	male	22	1	0	15160122	5.12	103	S
6	1	1	Mr. Wilson	male	35	0	0	15160122	5.12	103	S
7	0	3	Mr. Taylor	male	29	1	0	15160122	5.12	103	S
8	1	1	Mr. Moore	male	35	0	0	15160122	5.12	103	S
9	0	3	Mr. Jackson	male	22	1	0	15160122	5.12	103	S
10	1	1	Mr. White	male	35	0	0	15160122	5.12	103	S

titanicDat...

```
1 # import pandas as pd
2 # import numpy as np
3
4 ## Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6 # data = pd.get_dummies(data, columns=['Embarked'],
```

Average time: 0.145 s, Maximum time: 0.145 s, 1 out of 1 shown test case(s) passed

Test case 1 (145 ms) Debug

Expected output	Actual output
female: 233	female: 233
male: 109	male: 109
Name: Sex, dtype: int64	Name: Sex, dtype: int64
male: 468	male: 468
female: 81	female: 81
Name: Sex, dtype: int64	Name: Sex, dtype: int64

Terminal Test cases